

HTTP Protocol
Specification

for

TMR Current Sensing Terminal

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Date	Version	Memo
June 18, 2026	V0.0.1	Initial Version

Preface

This comprehensive specification is drafted to align software developers, hardware engineers and system integrators on end-to-end workflows for TMR Current Sensing terminals. All data retrieval operations are implemented via the standard HTTP GET method. Important note for the current development phase: only unsecure HTTP is supported, HTTPS is not available for use right now, and no authentication steps are needed to get valid data responses.

Currently supported GET list (Will extend if needed)

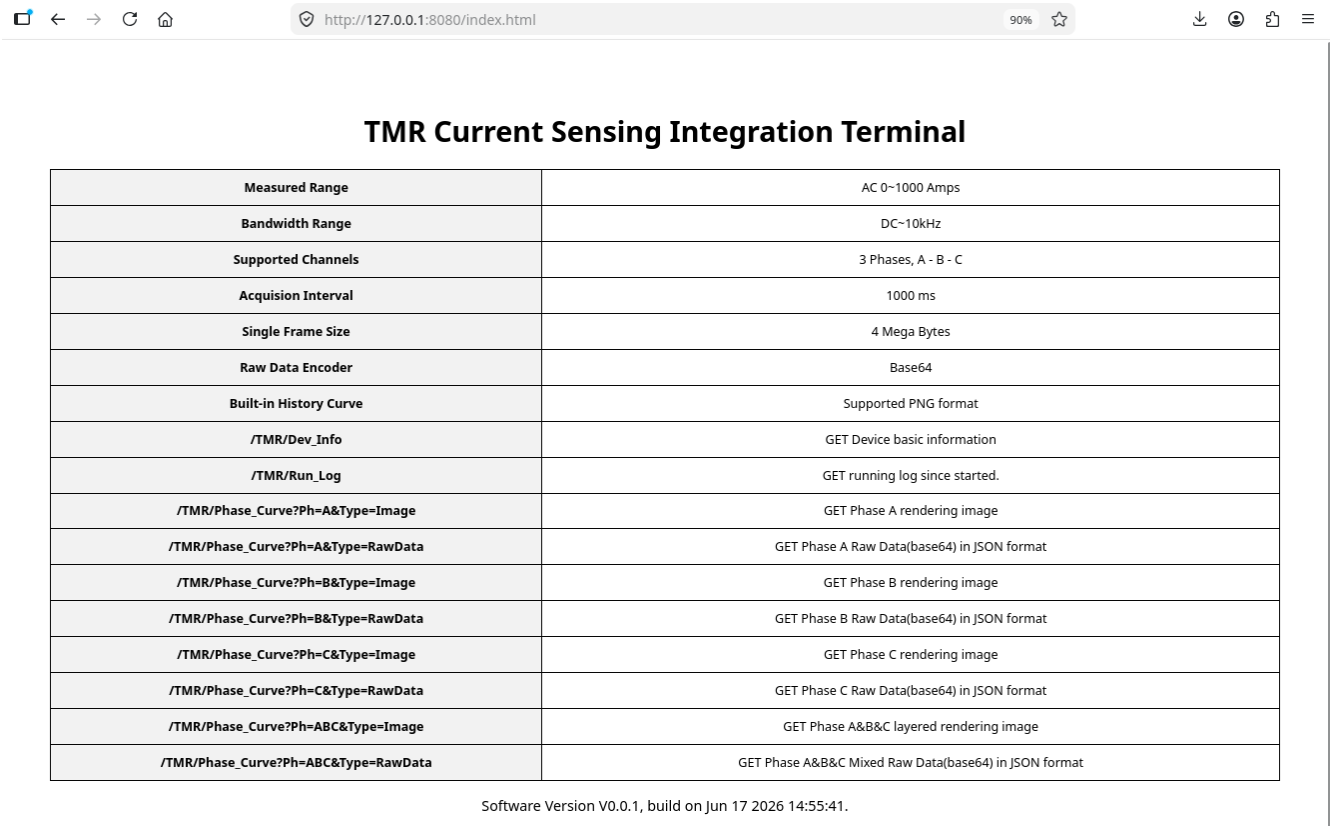
URL	Description
/index.html	The home page.
/TMR/Dev_Info	Get basic information of the terminal.
/TMR/Run_Log	Get running log since starts.
/TMR/Phase_Curve?Ph=A&Type=Image	Get Phase A rendering curve image.
/TMR/Phase_Curve?Ph=A&Type=RawData	Get Phase A raw data.
/TMR/Phase_Curve?Ph=B&Type=Image	Get Phase B rendering curve image.
/TMR/Phase_Curve?Ph=B&Type=RawData	Get Phase B raw data.
/TMR/Phase_Curve?Ph=C&Type=Image	Get Phase C rendering curve image.
/TMR/Phase_Curve?Ph=C&Type=RawData	Get Phase C raw data.
/TMR/Phase_Curve?Ph=ABC&Type=Image	Get Phase A+B+C layered rendering curve image.
/TMR/Phase_Curve?Ph=ABC&Type=RawData	Get Phase A+B+C mixed raw data.

/index.html

Description

Return the home page of the device. In this page, it displays the basic performance parameters of the terminal.

Screenshot



/TMR/Dev_Info

Description

To view the basic terminal details, issue a GET request to /TMR/Dev_Info, which returns a summary table of static device information.

Screenshot

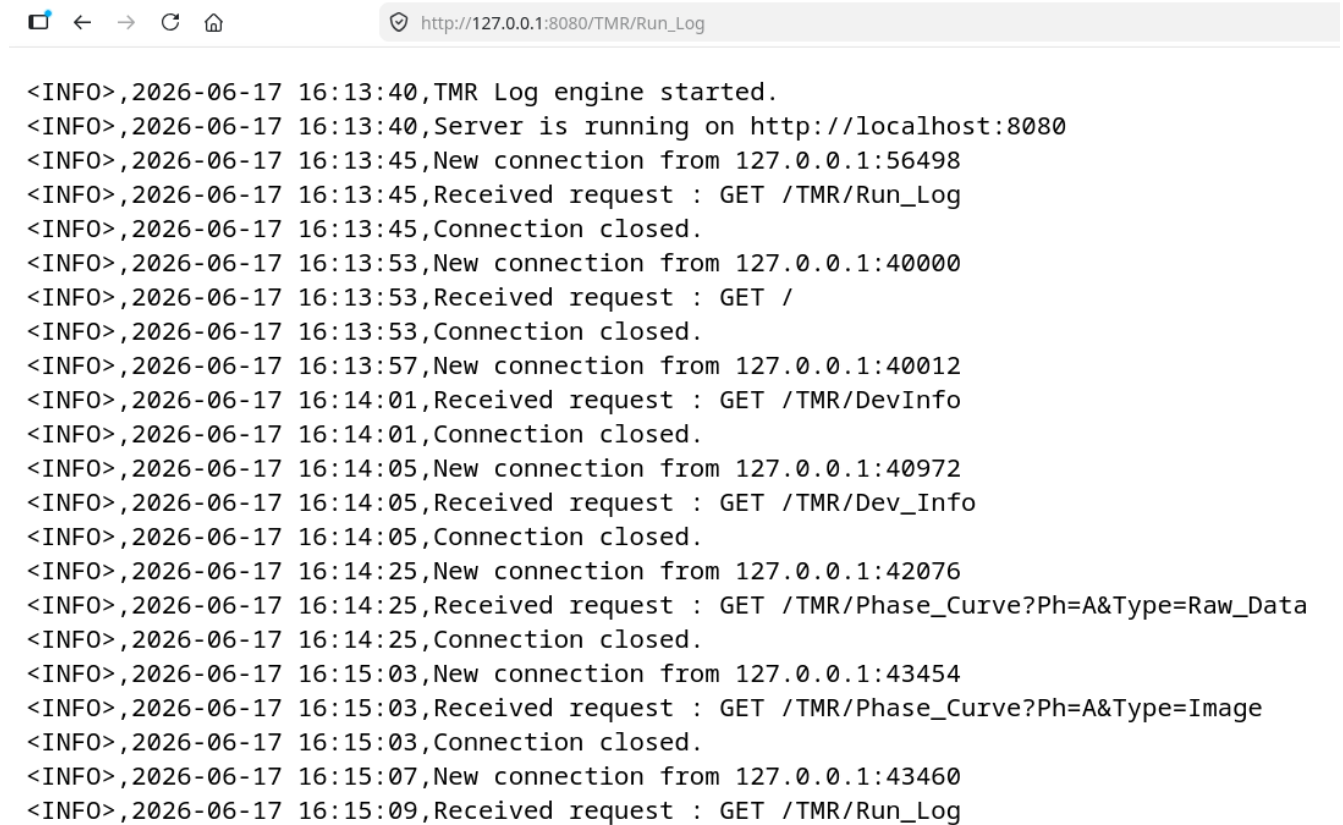


/TMR/Run_Log

Description

Send a GET request to /TMR/Run_Log to retrieve the full running log generated since device startup. The endpoint is designed to assist in rapid troubleshooting when errors occur.

Screenshot



The screenshot shows a web browser window with the address bar displaying `http://127.0.0.1:8080/TMR/Run_Log`. The main content area displays a log of server events in a monospaced font. The log entries are as follows:

```
<INFO>,2026-06-17 16:13:40,TMR Log engine started.
<INFO>,2026-06-17 16:13:40,Server is running on http://localhost:8080
<INFO>,2026-06-17 16:13:45,New connection from 127.0.0.1:56498
<INFO>,2026-06-17 16:13:45,Received request : GET /TMR/Run_Log
<INFO>,2026-06-17 16:13:45,Connection closed.
<INFO>,2026-06-17 16:13:53,New connection from 127.0.0.1:40000
<INFO>,2026-06-17 16:13:53,Received request : GET /
<INFO>,2026-06-17 16:13:53,Connection closed.
<INFO>,2026-06-17 16:13:57,New connection from 127.0.0.1:40012
<INFO>,2026-06-17 16:14:01,Received request : GET /TMR/DevInfo
<INFO>,2026-06-17 16:14:01,Connection closed.
<INFO>,2026-06-17 16:14:05,New connection from 127.0.0.1:40972
<INFO>,2026-06-17 16:14:05,Received request : GET /TMR/Dev_Info
<INFO>,2026-06-17 16:14:05,Connection closed.
<INFO>,2026-06-17 16:14:25,New connection from 127.0.0.1:42076
<INFO>,2026-06-17 16:14:25,Received request : GET /TMR/Phase_Curve?Ph=A&Type=Raw_Data
<INFO>,2026-06-17 16:14:25,Connection closed.
<INFO>,2026-06-17 16:15:03,New connection from 127.0.0.1:43454
<INFO>,2026-06-17 16:15:03,Received request : GET /TMR/Phase_Curve?Ph=A&Type=Image
<INFO>,2026-06-17 16:15:03,Connection closed.
<INFO>,2026-06-17 16:15:07,New connection from 127.0.0.1:43460
<INFO>,2026-06-17 16:15:09,Received request : GET /TMR/Run_Log
```

/TMR/Phase_Curve?Ph=A&Type=Image

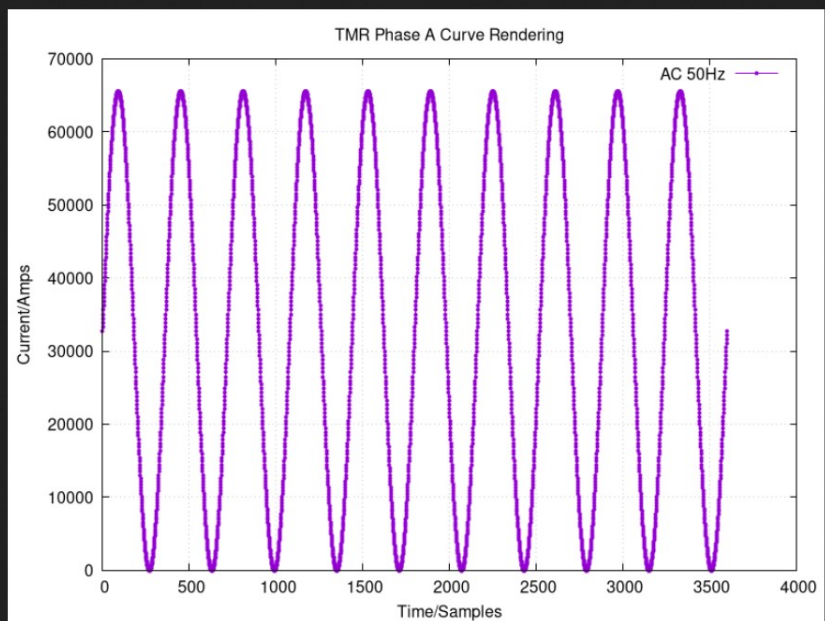
Description

Retrieve the rendered curve image for phase A. By default, the terminal outputs the curve in PNG format, and the total number of sampling points is configurable during the deployment phase.

Screenshot



http://127.0.0.1:8080/TMR/Phase_Curve?Ph=A&Type=Image

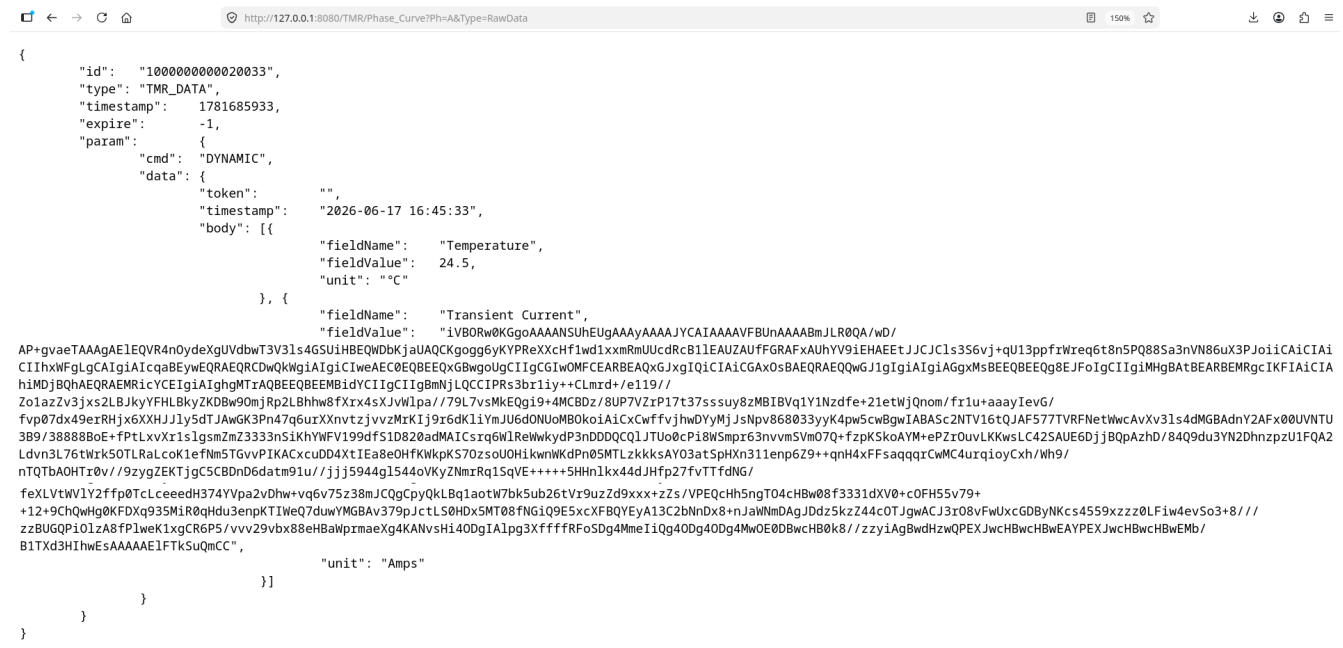


/TMR/Phase_Curve?Ph=A&Type=RawData

Description

Retrieve the raw data for phase A. The terminal acquires specified numbers of bytes of data, which is encoded via Base64 algorithm since HTML does not support binary data format. All data points are 4 bytes floating point values.

Screenshot



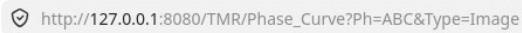
```
{
  "id": "100000000020033",
  "type": "TMR_DATA",
  "timestamp": 1781685933,
  "expire": -1,
  "param": {
    "cmd": "DYNAMIC",
    "data": {
      "token": "",
      "timestamp": "2026-06-17 16:45:33",
      "body": [
        {
          "fieldName": "Temperature",
          "fieldValue": 24.5,
          "unit": "°C"
        }, {
          "fieldName": "Transient Current",
          "fieldValue": "iVBORw0KGgoAAAANSUHEUgAAAYAAAAJYCAIAAAAFBUnAAAAABmJLR0QA/wD/AP+gvaeTAAAgAE1EQVR4n0ydeXgUVdbwT3V3ls4GSUjHBEQWDbKjaUAQCKgogg6yKYPRexXcHf1wd1xxmRmUcdRcB11EAUZAUFFGRAFXAUhYV9iEHAEETJJCJCLs356vj+qu13ppfxWreq6t8n5PQ885a3nVN86uX3PJoiiCAiCIA1CIIhXWfGLgCAIgiAIcqaBEyweQRAEQRCdWqKwgiAIgiCIweAEC0EQBEEQxGBwgoUgCIIgCGIw0MFCeARBEAQxGJxgIQiCIAiCGAxOsBAEQRAEQWgJ1gIgiAIgiAGgxMsBEEQBEEQg8JFoIgiIgiMHgBATBEARBEMRgcIKFIAiCIAhiMDjBQhAEQRAEMRiCYCEIgiAIghgMTraQBEQE8EEMBidYCIgCIIgBmNjLQCCIPRs3brliyy++CLmrd+/e119//Zo1azZv3jxs2LBJkyYFHLBkyZKDBw90mjRp2LBhhw8fXrx4sXJvWlpa//79L7vsMkEqgi9+4MCBDZ/8UP7VZrP17t37ssuy8zMbIBVq1Y1Nzdf+21etWjQnom/fr1u+aaayIevG/fvp07dx49erRHjx6XXHJJly5dTJAwG3Pn47q6uzXXnvtzjvvzMrKIj9r6dKliYmJU6d0NUoMB0koiAiCxCwffvjhwDyYmJJsNpv868033yyK4pw5cwBgwIABASc2NTV16tQJAF577TVRFNetWwcAvXv3ls4dMGBAdnY2AFx00UVNTU3B9/38888Boe+fPtLxvXr1s1gsmZm23333nS1KhYWFV199dfSID820adMAICsrq6W1ReWwkydP3nDDQDQCLJTUo0cP18W5mpr63nnvmSVm07Q+fzPKSkoAYM+ePZrOuvLKKwsLC42SAUE6DjjBQpAzhD/84Q9du3YNZDhnzpzUIFQA2Ldvn3L76tWrk50TLRaLcok1efNmSTGvvPIKACxcuD4xtIEa8e0HfKwkpKS70zsoUOHikwnWkDpn85MTLzkkksAY03atSPHXn311enp6Z9++qnH4xFFsaqqqrCwMC4uzqioyCxx/Wh9/nTQTbAOHTir0v//9zygZEKTjgC5CBdnD6datm91u//jjj5944g1544oVKyZNmrRq1SqVE++++5HHn1kx44dJHfp27fvTTFdNG/feXLVtWV1Y2ffp0TCLceedH374YVpa2vDhw+vg6v75z38mJCQgCpyQkLBq1aotW7bk5ub26tVi9uzZd9xxx+zZs/VPEQcHh5ngT04cHBw08f3331dXV0+c0FH55v79++12+9ChQwHq0KFDXq935MiR0qHdu3enpKTIWeQ7duwYMGBAV379pJctLS0HDx5MT08fNGiQ9E5xcXFBQYEyA13C2bNNDx8+nJawNmDagJDdz5kzZ44cOTJgwACJ3r08vFWUxcGDBYnKcs4559xzzz0LFiw4evSo3+8///zzBUGQP101zA8fP1weK1xgCR6P5/vvv29vbx88eHBAwprmaeXg4KANvsH140DgIAlpg3XfffrFo5Dg4MmeIiQg40Dg40Dg4MwOE0DBwcHB0k8//zzyiAgBwdHzwQPEXJwcHBwcHBwEAYPEXJwcHBwcHBwEmb/B1TXd3IhWesAAAAAE1FTkSuQmCC",
          "unit": "Amps"
        }
      ]
    }
  }
}
```

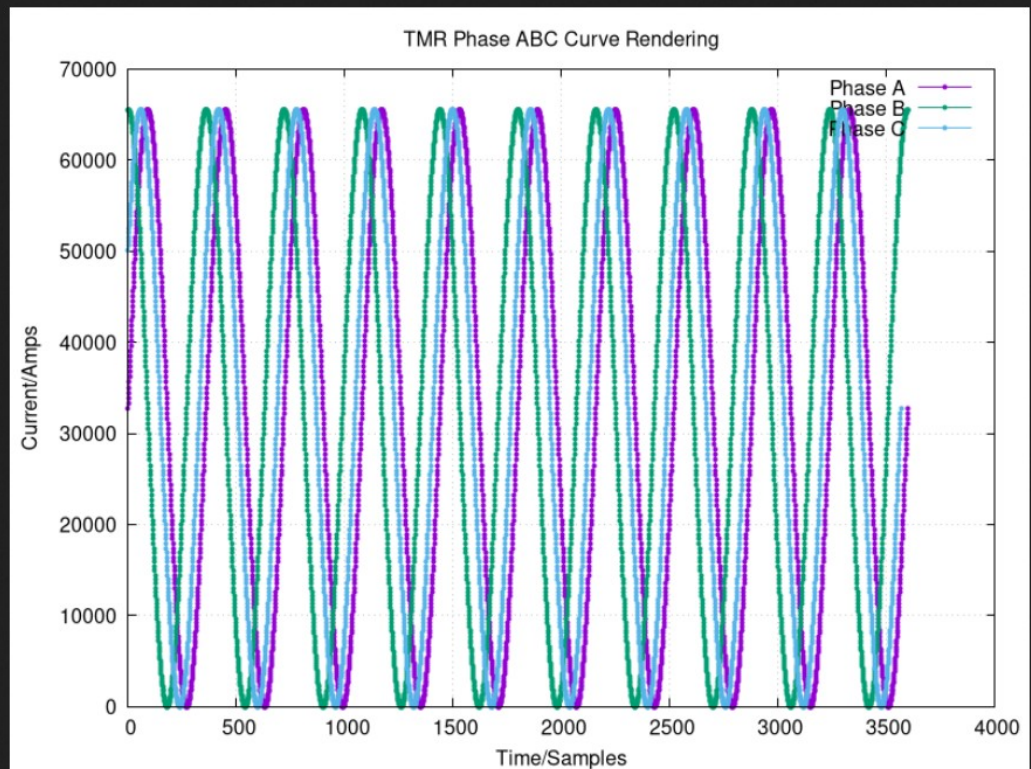
/TMR/Phase_Curve?Ph=ABC&Type=Image

Description

Retrieve the combined rendered curve image for phase A, B and C. By default, the terminal outputs the curve in PNG format, and the total number of sampling points is configurable during the deployment phase.

Screenshot

 http://127.0.0.1:8080/TMR/Phase_Curve?Ph=ABC&Type=Image



/TMR/Phase_Curve?Ph=ABC&Type=RawData

Description

Retrieve the raw data for phase A, B and C. The terminal acquires specified numberss of bytes of data, which is encoded via Base64 algorithm since HTML does not support binary data format. All data points are 4 bytes floating point values.

Screenshot

